

Response Letter with Annotations

Walter, D. & Ophir, Y. (2019). News Frame Analysis: An Inductive Mixed-method Computational Approach. *Communication Methods and Measures*, 13(4), pp. 248-266, <https://doi.org/10.1080/19312458.2019.1639145>.

Annotations

1. Purposes

- a. **Motivation (one sentence summary)** – Introduction of Analysis of Topic Model Networks (ANTMN), which combines topic modeling and network analysis to inductively identify and measure large amounts of text files for news frames.
- b. **Specific Rationale (several bullet points)**
 - i. Framing theory is difficult to operationalize in a standardized fashion in communication research
 - ii. There is a need for inductive methods that can identify frame elements and cluster them into frame packages based on textual evidence, rather than basic frames off of researcher assumptions
- c. **Research Questions**
 - i. **RQ1:** Can frames be identified inductively and validly from large corpora using computational techniques that combine topic modeling and network analysis?
 - ii. **IV/DV:** not applicable, as this is an exploratory methodological study to demonstrate.
 - iii. **Mediators/Moderators:** N/A

2. Methods

- a. **Research Design**
 - i. Three case studies to explore computational analysis for news frames
 - ii. Latent Dirichlet Allocation (LDA) topic modeling
 - iii. Network construction using cosine similarity
 - iv. Community detection algorithms
- b. **Sample**
 - i. Case Study 1: Political candidates
 - ii. Case Study 2: Foreign countries
 - iii. Case Study 3: Epidemics
- c. **Measures/Analysis**
 - i. Data processing
 1. Remove stop-words, converted to lowercase, removed punctuation/numbers
 2. Removed words appearing in >99% or <0.5% of documents
 3. No lemmatization or stemming
 - ii. Topic Modeling – LDA algorithm with Gibbs Sampling in R

- iii. Network construction – Topics as nodes, cosine similarity based on theta matrix for edges
- iv. Community detection – Walktrap algorithm, validated by Louvain, Spinglass, Fast-Greedy, Leading Eigenvector algorithms; sensitivity analysis conducted across platforms
- v. 2 independent coders analyzed topics using highest-loading words, FREX words, and representative documents

3. Results and Conclusions

a. Answers to RQ

- i. Case Study 1: Political Candidates
 - 1. 2 frames: (strategy and issue), both consistent with literature.
 - 2. Stability: nearly identical across all 5 community detection algorithms
- ii. Case Study 2: Foreign Countries
 - 1. Communities
 - a. conflict/violence
 - b. Diplomacy
 - c. Economy and sports
 - d. Culture
 - e. Cultural events
 - 2. Stable with variations in subdivisions
- iii. Case Study 3: Epidemics
 - 1. Communities
 - a. political/economic/cultural consequences and conflict
 - b. Disease invasion and containment
 - c. biological/medical facts
 - 2. Stable across algorithms

b. Author's Conclusions

- i. ANTMN identified frames across all three contexts
- ii. In established contexts (political campaigns), results matched existing theory with high validity
- iii. Emerging contexts method validated soft/hard power as frames and developed subcategories
- iv. Exploratory context method developed new frame structures
- v. Method can reduce researcher bias and reduce time and cost compared to manual coding, opens doors for future frame research

4. Discussion

a. Implications

- i. Supports emphasis framing over equivalency framing for news analysis
- ii. Demonstrates frames as communities of co-occurring linguistic elements

- iii. Suggests context-specific frames may be more common than generic ones
- iv. Scalable solution for big data analysis

b. Limitations

- i. Interpretation still requires qualitative/human analysis
- ii. Limited to emphasis frames
- iii. Irrelevant topics require manual removal
- iv. Each case study used limited examples
- v. Text-only frames

c. Future Directions

- i. Validation research
- ii. Cross-contextual research

5. Key Figures/Tables

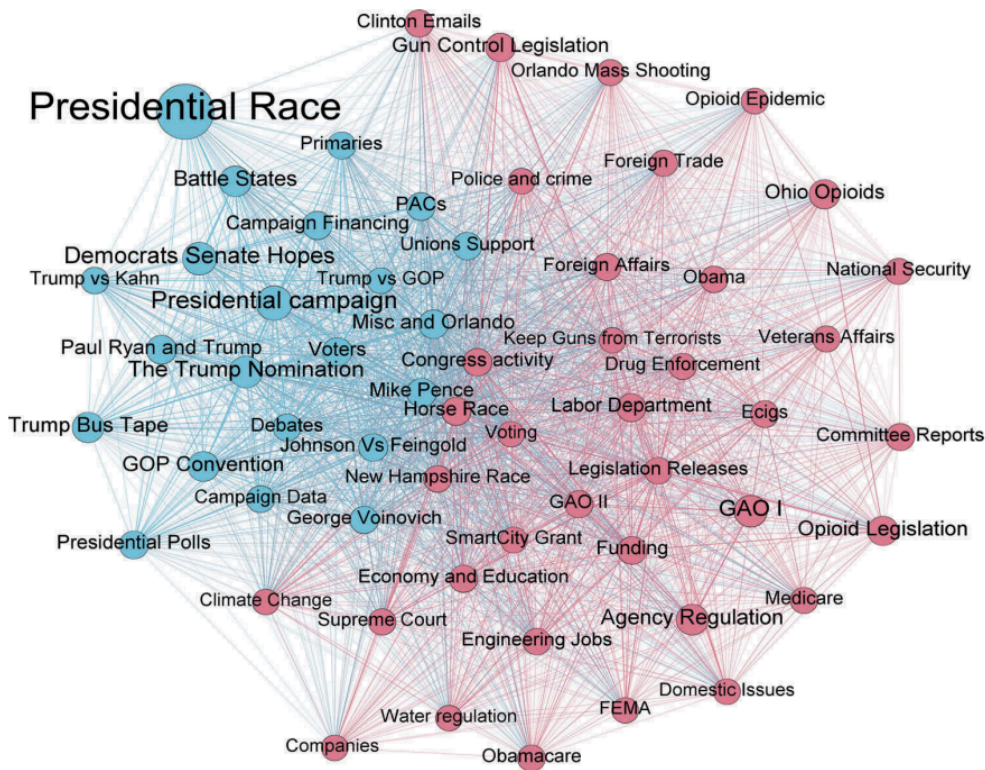
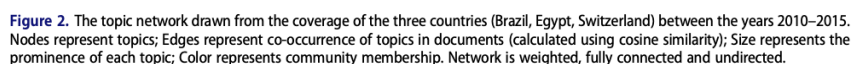
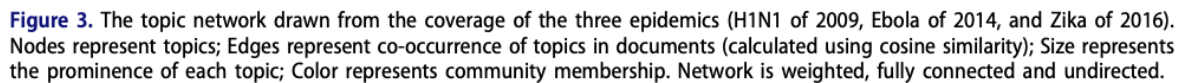


Figure 1. The topic network drawn from the coverage of the three Republican Senate candidates (Kelly Ayotte, Ron Johnson, and Rob Portman) in the 6 months prior to the 2016 elections. Nodes represent topics; Edges represent co-occurrence of topics in documents (calculated using cosine similarity); Size represents the prominence of each topic; Color represents community membership. Network is weighted, fully connected and undirected.



Response Letter

This study takes a computational approach to identify and measure news frames in large text corpora without researcher bias, as well as taking away the methodological constraints of manual coding. The researchers propose ANTMN (Analysis of Topic Model Networks) as a computational approach to combine topic modeling, network analysis, and community detection to inductively identify media frames.

The methodology has three prongs: (1) Latent Dirichlet Allocation (LDA) topic modeling identifies frame elements; (2) cosine similarity is calculated based on the data matrix of constructed networks to measure how frequently topics co-occur across documents; and (3) employing community detection algorithms to cluster topics into frame packages. The authors demonstrated this approach across three case studies: (1) Senate campaign coverage; (2) foreign country coverage; and (3) epidemic coverage.

One of the main strengths of this approach was its validity and stability across the various case studies and algorithms. Communities often remained largely consistent regardless of algorithmic choice. This method also solves the problem of analyzing large amounts of articles across studies, which would be virtually impossible to code manually. The study is also reproducible and transparent in methods, which offer future research directions as well as guidance in the development of computational methods for news framing study.

However, while the study cuts down a lot of effort for researchers, as well as potential researcher bias in qualitative analysis, human interpretation is still fundamental to the methodology. Some subjectivity remains in the methodology as researchers must label topics and communities, and remove irrelevant articles and boilerplate topics. This dependence on researcher judgment does leave room for questions about validity and bias in the research. Materially, the biggest weakness of this study is that the approach is limited to emphasis framing in text, and cannot be used to evaluate equivalency frames, or visual cues.

This method ultimately reevaluates how news framing research can be done in the wake of developing computational methodology. This invokes the following questions for study:

1. If human interpretation is still required in the computational analysis, is the bias really reduced? Have we just relocated the bias to a different stage of the analysis?
2. Could this research be extrapolated to computational analysis of survey or experimental research?
3. How could cross-contextual research benefit from this methodology?